ZUSE Automat Agent: Empirical Law Discovery in Elementary Cellular Automata

Abstract

We present ZUSE Automat Agent (ZUSE), a deterministic, policy-driven discovery loop for elementary cellular automata (ECA) that builds an empirical atlas of cycle laws, world families, and basin fragility without a language model in the discovery loop. The pipeline combines a fixed seven-law evaluator, a dedup-gated observer stack, and persistent multi-seed world records, and is applied to a 20-world sample spanning ECA rules, Conway's Game of Life patterns, and synthetic controls.

The seven laws — *velocidad_constante*, *periodicidad*, *densidad_estable*, *tipo_unico*, *complejidad_alta*, *frontera_temporal*, and *temporal_scale_stability* — are calibrated empirically (*frontera_temporal* upper bound 0.4352; *temporal_scale_stability* threshold 19.03, decision-tree accuracy 0.908). Laws are separated into two groups: structure-observer laws that depend on the full observer stack, and frame-metric laws that depend only on aggregate frame statistics.

Key results: (1) *frontera_temporal* is not intrinsically rare — it activates in 38 of 256 ECA rules under at least two of three seeds, and in 17 of 256 under all three; (2) the 20-world atlas reveals seven dynamic categories (*frontera-rich-estable*, *periodicidad-global*, *oscilador-local*, *multiregimen-productivo*, *multiregimen-escala-dependiente*, *noise-bounded*, *sin-evidencia-multiregimen*), finer than Wolfram's four-class taxonomy; (3) one-bit IC fragility spans from perfectly stable basins (`rule_208/209`, `f_total = 0.000`) to exact fixture disruption (`life_blinker`, `f_total = 1.000`), with `rule_108` as the ECA outlier (`f_total = 0.992`, `f_gap = 0.945`). These cases separate three mechanistically distinct regimes: productive basin switching, noise-boundary crossing, and quiescent-background activation; (4) an exhaustive protocol over 128 quiescent ECA rules and 502 non-zero IC words per rule confirms `rule_108` as the unique ECA rule producing stationary local period-2 oscillators; (5) designed periodic ICs activate production `periodicidad` in 207 of 256 ECA rules, showing that the law is IC-family sensitive rather than inaccessible; and (6) a controlled single-bit experiment demonstrates that ECA frames are translation-invariant while the observer/dedup pipeline is not, separating physical law from measurement artifact; and (7) three progressive background-oscillator sweeps reveal that the local oscillator landscape is strongly background-conditioned. Under quiescent zero background, only `rule_108` (stationary, T=2) and eight rules (moving, T=2, speed 1 cell/step) produce local oscillators. Under non-zero periodic backgrounds of template length 1, 2, and 4 (1,927,680 runs), the landscape expands to 30 stationary and 36 moving rules, introducing period-4 oscillators and speed-0.5 gliders. Under 30 primitive length-8 binary backgrounds (3,855,360 runs), 23 further rule/type pairs appear, the observed period extends to T=15, and speed 2/3 cell/step is observed for the first time. Phase sensitivity is detected in all 10 sampled rules from the length-8 sweep; a strict IC/background co-translation test confirms exact physical equivariance in 80/80 runs after correcting a cyclic-boundary artifact in `linear_shape`. The `T=15` family is confined to the reflection-symmetric, black/white-conjugate pair `rule_73/rule_109`, locks at five times the background temporal period, and persists through step 900 in all 20 minimal rule/background representatives. Sampling the localized defect once per background period reveals a minimal five-state cycle under `F^3` in 20/20 representatives, establishing the computational mechanism of the measured 5:1 locking ratio. The induced defect rule `delta_f(b,d)=f(b XOR d) XOR f(b)` further yields an exact analytical conjugation identity between `rule_73` and `rule_109`; exhaustive checks reject a fixed sparse truth-table support as the universal source of the cycle.

Every result is reproducible from deterministic scripts with no stochastic components in the discovery loop.

1. Introduction

Elementary cellular automata are among the simplest systems known to exhibit complex behavior. A radius-1, binary, one-dimensional CA is fully specified by a single integer from 0 to 255, yet even within this minimal space, Wolfram's empirical taxonomy finds four qualitatively distinct dynamic classes: uniform, periodic, locally chaotic, and complex. Cook's proof that Rule 110 supports universal computation establishes that complexity in ECA is not merely visual: it has computational consequences.

Two questions remain largely open after Wolfram's program. First, the taxonomy is qualitative and coarse: all complex rules fall into Class 4 regardless of their intra-class differences in structure type, periodicity, fragility, or scale behavior. Second, the boundary between the dynamics of the rule and the properties of the measurement instrument is rarely made explicit. When an observer reports that a run contains gliders, it conflates the CA physics with the heuristic that defined the glider label.

ZUSE Automat Agent addresses both questions through a deterministic, policy-driven discovery loop. The agent runs ECA worlds, applies a fixed stack of heuristic observers, evaluates seven binary cycle laws, and stores multi-seed evidence in persistent world records. No language model participates in the discovery loop: law proposals, world selection, and evidence evaluation are all deterministic. Language-model assistance is restricted to post-run interpretation and documentation.

The result is an empirical atlas of 20 worlds with seven operational categories, measured fragility along two axes (`f_total` and `f_core`), and explicit characterization of two observer artifacts. The atlas is not a new taxonomy of Wolfram's classes; it is a finer-grained, evidence-based map of a 20-world sample that separates cycle-level laws, world-level regimes, and pipeline behavior.

1.1 Contributions

We make the following contributions:

1. **ZUSE Automat Agent** — A deterministic, policy-driven discovery loop for ECA that accumulates multi-seed law evidence across worlds without symbolic regression or LLM guidance in the loop. The agent combines persistent world-record history, a dedup-gated observer stack, and a seven-law evaluator into a single reproducible pipeline.

2. **A seven-category empirical atlas of 20 worlds** — We classify 20 worlds spanning ECA rules, Conway's Game of Life patterns, and synthetic controls into seven operational categories (*frontera-rich-estable*, *periodicidad-global*, *oscilador-local*, *multiregimen-productivo*, *multiregimen-escala-dependiente*, *noise-bounded*, *sin-evidencia-multiregimen*) using law coverage, signature diversity, and fragility. The atlas extends Wolfram's four-class taxonomy by capturing intra-class structure, scale-dependent silencing, and negative-control regimes.

3. **A two-dimensional fragility framework** — We measure `f_total` and `f_core` separately, defining `f_gap = f_total - f_core` as a quantitative measure of secondary-law churn. Four distinct mechanisms are identified: stable basin (`rule_208/209`, `f_total = 0.000`), productive basin switching (`rule_137`, `f_gap = 0.318`), noise-boundary fragility (`rule_54`, `f_core = 0.677`), and quiescent-background activation (`rule_108`, `f_gap = 0.945`).

4. **`rule_108` as the unique stationary local-period-2 ECA oscillator** — Under an exhaustive protocol (128 quiescent ECA rules, 502 non-zero IC words per rule, span <= 32, period <= 16), `rule_108` is the only ECA rule that produces stationary local period-2 oscillators. The motif `#.# <-> ###` follows algebraically from `f(0,1,0) = f(1,0,1) = 1` and `f(1,1,1) = 0`, and the rule's left-right symmetry (`f(l,c,r) = f(r,c,l)`) explains why the oscillator does not drift.

5. **A measured separation between ECA dynamics and observer artifacts** — `rule_54` single-bit-IC frames are provably translation-invariant (confirmed by frame identity after shift normalization), while observer dedup counts range from 15 to 24 across IC positions. This non-equivariance is characterized as a pipeline property: absolute structure counts depend on IC context, but law signatures remain stable.

2. Related Work

ZUSE sits at the intersection of three bodies of prior work: empirical ECA taxonomy, automated law discovery, and LLM-based scientific agents. It is related to each tradition but positioned differently from all three: it extends ECA taxonomy inward (intra-class rather than inter-class), it inverts the discovery framing of symbolic regression systems (fixed evaluators rather than generative hypotheses), and it excludes the LLM from the loop rather than centering it.

2.1 ECA taxonomy and complexity

Wolfram's systematic study of elementary cellular automata established the canonical four-class taxonomy: Class 1 (uniform), Class 2 (periodic), Class 3 (chaotic), and Class 4 (complex) [Wolfram2002]. This taxonomy is qualitative and based on visual inspection of space-time diagrams. ZUSE extends it by measuring intra-class structure: two Class-4 rules (`rule_137` and `rule_54`) differ not only in fragility magnitude but in fragility mechanism, a distinction the four-class taxonomy does not capture.

Cook's proof that Rule 110 supports universal computation [Cook2004] established that ECA complexity has computational consequences beyond visual appearance. But computational class is a coarse lens: `rule_110` and `rule_54` are both Class 4, yet ZUSE finds that their fragility mechanisms are qualitatively different — productive basin switching versus noise-boundary crossing. `rule_110` appears in the ZUSE atlas as a `multiregimen-productivo` world with `f_total = 0.323` and a stable frontera signature. The computational proof characterizes what `rule_110` *can* compute; ZUSE characterizes what it *typically does* under random initialization.

2.2 Automated scientific discovery

AI Feynman [Udrescu2020] demonstrated symbolic regression over physical datasets, recovering known equations from data with interpretable structure. The contrast with ZUSE is deliberate: AI Feynman uses neural networks to propose candidate laws from continuous-variable data, while ZUSE applies fixed binary evaluators to discrete CA dynamics and accumulates evidence without a generative component. ZUSE is not a symbolic regression system; it is a policy-driven measurement pipeline whose outputs are law signatures, not formulas.

More broadly, systems such as Eureqa [Schmidt2009] and recent LLM-based discovery agents frame discovery as hypothesis generation followed by verification. ZUSE inverts this framing: laws are fixed a priori, and the discovery consists of finding which worlds satisfy them and under what conditions. This makes every accepted law signature verifiable from deterministic scripts, at the cost of not proposing new laws automatically.

2.3 ZUSE as evidence engine, not LLM scientist

Recent work on LLM-based scientific agents (e.g., The AI Scientist [Lu2024]) demonstrates that language models can propose hypotheses, design experiments, and write papers with minimal human intervention. ZUSE occupies a different position in this space:

the language model is explicitly excluded from the discovery loop and restricted to post-run interpretation and documentation.

This separation is a design choice, not a limitation. It means that the atlas findings are fully reproducible from the deterministic loop code, and that language-model involvement can be audited at the documentation layer without contaminating the empirical results. The cost is that ZUSE cannot propose new laws; the benefit is that every accepted law has a transparent, non-generative provenance.

3. System: ZUSE Automat Agent

ZUSE Automat Agent is a deterministic discovery loop over cellular automaton worlds. A world is a simulator plus an initial-condition protocol and a time window. For ECA worlds, the simulator is the standard binary radius-1 update rule with periodic boundary conditions. The agent runs a world, computes frame metrics, extracts candidate structures, evaluates a fixed set of laws, updates world-level history, and chooses the next action through a transparent policy.

3.1 Inputs and outputs

A single agent cycle takes as input:

- **World identifier**: the ECA rule number (`0..255`) or a named synthetic world.
- **IC protocol**: either a random seed for standard runs or a designed IC vector for controlled experiments.
- **Width and steps**: fixed integers, typically `width = 64` and `steps = 24..200`, depending on the world's scale protocol.

The outputs of a single cycle are:

- **Frame metrics**: density mean, entropy mean, temporal transition rate, gzip ratio, and mutual information mean.
- **Analysis status**: `ok` or `ruido_no_analizable` after the noise gate.
- **Structure records**: raw `Estructura` outputs with type labels and span information, plus `dedup_structure_count`.
- **Law signature**: frozenset of accepted law names, or the empty set if the run is noise-gated.

3.2 Loop structure

The loop has five layers:

1. **Simulation.** ECA frames are generated from explicit initial conditions with fixed `width`, `steps`, and `seed` or with designed ICs for controlled experiments. The simulator itself is not learned.
2. **Frame metrics.** Each run is summarized by density, entropy, temporal transition rate, gzip compressibility, and temporal mutual information. These features support both individual laws (`complejidad_alta`, `frontera_temporal`, `temporal_scale_stability`) and later meta-analysis.
3. **Observers and deduplication.** A stack of heuristic observers converts frame histories into `Estructura` records with type labels such as `glider`, `bloque`, and `oscilador`. The raw observer outputs are intentionally kept for audit, while `deduplicate_structures` estimates the number of physical structures. The production noise gate uses `dedup_structure_count > 40`.
4. **Cycle-law evaluation.** Seven laws are evaluated on each analyzable run. The result is a law signature: the set of accepted laws for that cycle. Noise-gated runs skip law evaluation rather than forcing a low-confidence signature.
5. **Policy and memory.** The agent stores a persistent `WorldRecord` per world and chooses the next action after each cycle.

3.3 State: WorldRecord

Each world maintains a `WorldRecord` with the following fields relevant to atlas construction:

field	description
`visit_count`	total cycles run on this world
`scores`	per-cycle scores used by the policy
`noise_count` / `noise_fraction`	count and fraction of noise-gated cycles
`law_signatures`	list of accepted law signatures as frozensets
`unique_law_signature_count`	count of distinct non-empty signatures
`non_empty_signature_visit_count`	visits where at least one law was accepted
`law_signature_diversity`	unique non-empty signatures divided by non-empty visits, reported after at least five non-empty visits
`peak_signature_diversity`	maximum clean diversity observed so far
`has_multiregime_evidence`	monotone boolean, set once peak diversity exceeds `0.5` under low noise
`params_tried`	tested `(steps, width, law_signature)` tuples
`max_ok_steps` / `first_noise_steps`	scale boundary diagnostics

The important design choice is that empty signatures are retained for audit but excluded from diversity. A world is not multi-regime merely because it alternates between laws and silence; multi-regime evidence requires multiple non-empty law signatures.

3.4 Journal

Every cycle result is appended to a JSONL journal (`outputs/experiments_*/journal_*.jsonl`). Each line is a self-contained JSON record with cycle identifier, world identifier, steps, width, frame metrics, analysis status, structure counts, law signature, action taken, and the previous WorldRecord state visible to the policy at decision time. The journal is the primary reproducibility artifact: the atlas in Section 5, the fragility measurements in Section 6, and the case studies in Section 7 are all derived from journal queries and controlled follow-up scripts.

3.5 Policy

The policy selects the next action from four options:

- **Vary seed** (`repeat_vary_seed`): run the same world with a new IC. Used when the world has a new law signature or confirmed multi-regime evidence and the current cycle is productive.
- **Increase scale** (`increase_steps`): raise `steps` to test scale-dependent behavior. Used when the current world produces analyzable signal and has not reached a known noise boundary.
- **Change world** (`change_world`): move to the next world. Used when the current world is noise-bounded, reaches a known noise boundary, has exhausted repeats, or converges to unproductive silence at maximum scale.
- **Stop by exhaustion**: after the requested cycle budget, persist state and journal artifacts.

The policy has no learned parameters. Its thresholds are fixed constants or explicit guards in code: `dedup_structure_count > 40` for the noise gate, signature diversity `> 0.5` for multi-regime evidence, `noise_fraction < 0.20` for clean diversity, and at least five non-empty visits before diversity is reported.

3.6 Non-generative design

The agent is deliberately non-generative inside the loop. No LLM proposes laws, selects worlds, or evaluates a cycle. Symbolic regression was used only outside the loop for calibration and analysis; it is not part of the online discovery policy. The LLM-assisted work reported here occurs after runs are complete: it helps design follow-up experiments, interpret artifacts, and write documentation. This separation is important because every accepted law signature in the atlas can be reproduced from deterministic scripts.

4. Seven Cycle Laws

The seven laws are the primary evidence units linking raw ECA frames to the world categories, fragility scores, and observer artifacts reported in Sections 5-8. Each law is evaluated per run — one world, one initial condition, one step count — once the observer and dedup pipeline reports `analysis_status = ok`. The output is binary: accepted or rejected. Law signatures are frozensets of accepted law names.

4.1 Design rationale

The seven laws were chosen to span distinct aspects of CA behavior using measurements available from a single fixed-length run. They divide into two groups by input type:

- **Structure-observer laws** (`velocidad_constante`, `periodicidad`, `densidad_estable`, `tipo_unico`): depend on the output of the heuristic observer stack. They can only fire if the run is analyzable (`analysis_status = ok`) and the observers detect at least one structure.

- **Frame-metric laws** (`complejidad_alta`, `frontera_temporal`, `temporal_scale_stability`): depend only on summary statistics computed directly from the frame array, without reference to individual structures.

This split is intentional. Frame-metric laws are symmetry-agnostic: they fire regardless of where structures are in the lattice, how many there are, or whether the observers identify them correctly. Structure-observer laws are richer but carry the observer's heuristic assumptions. Section 8 reports two cases where structure-observer laws produce artifacts that frame-metric laws do not.

All laws produce binary output (accepted / rejected). The choice of binary rather than continuous output is also intentional: it makes signatures comparable across runs and worlds without requiring score normalization, and it forces explicit calibration of each threshold.

4.2 Formal criteria

#	Law	Inputs	Criterion	Constants
1	`velocidad_constante`	Position tracks of moving structures	At least 50% of moving tracks (`velocity > 0.05` cells/step) have linear `x(t)` with normalized residual `< 0.15`	-

#	Law	Inputs	Criterion	Constants
2	`periodicidad`	Structure type list	At least one structure classified as `oscilador`	-
3	`densidad_estable`	Frame density time series	Coefficient of variation `CV = sigma(rho) / mu(rho) < 0.15`	-
4	`tipo_unico`	Structure type set	Exactly one structure type present	-
5	`complejidad_alta`	Frame metrics	`entropy_mean > 0.80` and `transition_rate > 0.25`	-
6	`frontera_temporal`	Frame metrics	`entropy_mean > 0.80` and `0.28 < transition_rate < 0.4352`	upper threshold calibrated 2026-05-24
7	`temporal_scale_stability`	Frame metrics + steps	`temporal_load = steps * gzip_ratio / transition_rate < 19.03`	threshold calibrated 2026-05-24

`temporal_scale_stability` rejects any run with `transition_rate = 0` (quiescent or static configurations), since temporal load is undefined (`infinity`).

4.3 Calibrated constants

Neither threshold can be derived analytically: the boundary between organized frontier dynamics and pure chaos has no closed form in ECA. Both constants were set empirically on real ECA runs and are valid within the atlas protocol (`width = 64`, `steps` roughly `24..200`).

The `frontera_temporal` upper threshold `0.4352` is the midpoint between the maximum `transition_rate` observed for `rule_110` (`0.4147`) and the minimum for `rule_30` (`0.4557`) across six canonical seeds at `steps = 24`, `width = 64`.

The `temporal_scale_stability` threshold `19.03` was fit on `datasets/fase2c_v3.csv` (120 ECA scale samples). A decision tree at `max_depth = 4` achieved accuracy `0.908`, precision `0.886`, and recall `0.954` on the `analysis_ok` label.

4.4 Caveats

`tipo_unico` is an observer-dependent exploratory signal, not a mirror-invariant physical property. Fase 6b showed that `rule_110` and `rule_124` are left-right mirrors of each other with identical dynamics, yet `tipo_unico` can fire asymmetrically depending on orientation. `tipo_unico` is retained in the atlas for its exploratory value but should not be used as evidence of physical asymmetry.

`frontera_temporal` and `temporal_scale_stability` both depend on `transition_rate`. Fase 4a and later tree analyses identify transition rate as the main discriminator separating organized frontier dynamics from pure chaos or static order. Other metrics (`density_mean`, `gzip_ratio`, `mutual_info_mean`) are useful context features but should not be treated as independent causal evidence without ablation.

These caveats do not weaken the atlas: they clarify which signals reflect physical ECA dynamics and which reflect the current observer design. Section 8 returns to both artifacts with controlled experiments.

4.5 Law signatures and the atlas

A law signature is a frozenset of accepted law names for one run. The empty frozenset is valid and indicates a run that passed the noise gate but accepted no law. Law signatures are the unit of evidence in the atlas: the world categories in Section 5 are defined by how signatures distribute across seeds and scales, and the fragility measurements in Section 6 count how often signatures change under perturbation.

`frontera_temporal` is a proper subset of `complejidad_alta` by construction (it adds the upper bound on transition rate). Any run that accepts `frontera_temporal` also accepts `complejidad_alta`; the converse is not required. This containment is visible in the law coverage matrix: every ✓ in the `frontera_temporal` column co-occurs with a ✓ in the `complejidad_alta` column.

5. World Atlas: 20 Worlds and Dynamic Categories

The atlas is derived from `outputs/world_taxonomy/law_map.md`. It contains 20 worlds: ECA rules, designed synthetic controls, and Life-like controls. Each world is summarized by law coverage, non-empty visit ratio, noise ratio, signature diversity, mean law count, dominant signature, and measured fragility where available.

The atlas is not a score table. A high `mean_laws` value, high signature diversity, and low fragility mean different things. The taxonomy therefore separates five positive dynamic families from two bookkeeping categories: `noise-bounded` for worlds stopped by the dedup gate, and `sin-evidencia-multiregimen` for controls or worlds without sufficient evidence for one of the positive families.

5.1 Category definitions

category	operational signal	representative worlds
`frontera-rich-estable`	low signature diversity, high stable law richness (`mean_laws >= 4.0`)	`rule_46` , `rule_208` , `rule_209`
`periodicidad-global`	global period-2 behavior; `periodicidad` in nearly all non-empty visits	`rule_51`
`oscilador-local`	bounded local period-2 structure on a quiescent background	`rule_108`
`multiregimen-productivo`	multiple non-empty law signatures with productive visits	`rule_18` , `rule_54` , `rule_109` , `rule_110` , `rule_124` , `rule_137`
`multiregimen-escala-dependiente`	real signature diversity but most high-scale visits become analyzable silence	`rule_90`
`noise-bounded`	pre-law failure under the deduplicated structure gate	`rule_30` , `rule_150`
`sin-evidencia-multiregimen`	no sufficient evidence of multi-regime or stable-rich behavior in the current protocol	`life_blinker` , `life_block` , `life_glider` , `synthetic_bloque` , `synthetic_glider` , `synthetic_oscilador`

world	category	mean_laws	peak_diversity	f_total	f_core
`rule_208`	`frontera-rich-estable`	6.000	0.167	0.000	0.000
`rule_209`	`frontera-rich-estable`	6.000	0.167	0.000	0.000
`rule_46`	`frontera-rich-estable`	5.833	0.333	0.031	0.031
`rule_51`	`periodicidad-global`	4.500	0.333	0.193	0.000
`rule_108`	`oscilador-local`	2.000	0.167	0.992	0.047
`rule_90`	`multiregimen-escala-dependiente`	0.500	0.600	0.172	0.000
`rule_110`	`multiregimen-productivo`	2.727	0.600	0.323	0.198
`rule_124`	`multiregimen-productivo`	2.167	0.600	0.224	0.083
`rule_109`	`multiregimen-productivo`	2.000	0.667	0.307	0.307
`rule_18`	`multiregimen-productivo`	2.308	0.800	0.349	0.135
`rule_137`	`multiregimen-productivo`	2.867	0.833	0.630	0.312
`rule_54`	`multiregimen-productivo`	1.917	0.800	0.714	0.677
`rule_30`	`noise-bounded`	1.100	0.000	0.021	0.021
`rule_150`	`noise-bounded`	0.750	0.000	0.023	0.023
`life_blinker`	`sin-evidencia-multiregimen`	3.000	0.200	1.000	1.000
`life_block`	`sin-evidencia-multiregimen`	2.000	0.200	0.016	0.016
`life_glider`	`sin-evidencia-multiregimen`	2.357	0.333	0.032	0.032
`synthetic_bloque`	`sin-evidencia-multiregimen`	2.000	0.200	n/a	n/a
`synthetic_glider`	`sin-evidencia-multiregimen`	3.167	0.400	n/a	n/a
`synthetic_oscilador`	`sin-evidencia-multiregimen`	2.286	0.400	n/a	n/a

This classification is intentionally operational. A world can be reclassified if a wider protocol produces different evidence; the atlas records what the current deterministic protocol has measured.

The same principle applies to `frontera_temporal` candidates. Fase 20a found many additional ECA rules that are rich in `frontera_temporal` at the fixed sweep scale (`steps = 24`), but Fase 20b showed that the strongest four new candidates do not remain `frontera-rich-estable` under long-journal policy scaling. Fase 20c therefore treats `frontera-short-scale` as a candidate tier, not as an atlas-grade category. The atlas promotes worlds only when short-scale richness survives the broader validation protocol.

5.2 Law coverage

The law coverage matrix uses seven columns:

```
velocidad_constante
periodicidad
densidad_estable
tipo_unico
complejidad_alta
frontera_temporal
temporal_scale_stability
```

Each cell has one of four states: accepted in the dominant signature or in at least half of non-empty visits (`✓`), observed but below half (`·`), never observed in non-empty visits (`-`), or unknown because no non-empty visits exist (`?`).

Cell states:

- `✓`: law appears in the dominant signature or in at least 50% of non-empty visits.
- `·`: law appears in at least one non-empty visit but in less than 50%.
- `-`: non-empty visits exist and the law never appears.
- `?`: no non-empty visits.

world	vel	per	den	tipo	compl	front	tss
`life_blinker`	-	✓	✓	✓	-	-	-
`life_block`	-	-	✓	✓	-	-	-
`life_glider`	·	-	✓	✓	-	-	-
`rule_108`	-	✓	-	✓	-	-	-
`rule_109`	-	-	✓	·	✓	✓	✓
`rule_110`	✓	-	✓	-	✓	✓	·
`rule_124`	·	-	✓	-	✓	✓	✓
`rule_137`	·	-	✓	·	✓	✓	✓
`rule_150`	-	-	✓	-	✓	-	✓
`rule_18`	✓	-	-	✓	✓	-	✓
`rule_208`	✓	-	✓	✓	✓	✓	✓
`rule_209`	✓	-	✓	✓	✓	✓	✓
`rule_30`	-	-	✓	-	✓	-	✓
`rule_46`	✓	-	✓	✓	✓	✓	✓
`rule_51`	-	✓	✓	✓	✓	-	✓
`rule_54`	✓	-	·	·	✓	-	✓
`rule_90`	·	-	·	-	·	-	✓
`synthetic_bloque`	-	-	✓	✓	-	-	-
`synthetic_glider`	✓	-	✓	✓	-	-	·
`synthetic_oscilador`	-	✓	-	✓	-	-	·

The matrix reveals three broad patterns:

1. **Synthetic and Life-like controls validate observer semantics.** `life_blinker` and `synthetic_oscilador` activate `periodicidad`; block-like worlds activate `densidad_estable` and `tipo_unico`; synthetic gliders activate `velocidad_constante`.

2. **Class-4 and frontier worlds separate into distinct families.** `rule_137`, `rule_110`, `rule_124`, `rule_109`, and `rule_54` are multi-regime worlds with two or three dominant laws. By contrast, `rule_46`, `rule_208`, and `rule_209` activate six of seven laws with low diversity.

3. **`periodicidad` is IC-family sensitive.** Under random ICs it appears in designed controls, in the global complement rule (`rule_51`), and in the local oscillator (`rule_108`), but not in the complex frontier worlds. Under explicitly periodic ICs, however, Fase 21a finds production `periodicidad` in `207/256` ECA rules. The law is therefore not dead or ECA-inaccessible; it is controlled by the IC family. This is why Section 7 treats `rule_108` separately rather than folding it into ordinary stable-rich behavior.

5.3 Key atlas rows

The following rows anchor the category structure:

world	category	mean_laws	peak_diversity	dominant signature
`rule_208`	`frontera-rich-estable`	`6.000`	`0.167`	`velocidad_constante + densidad_estable + tipo_unico + complejidad_alta + frontera_temporal + temporal_scale_stability`
`rule_209`	`frontera-rich-estable`	`6.000`	`0.167`	`velocidad_constante + densidad_estable + tipo_unico + complejidad_alta + frontera_temporal + temporal_scale_stability`
`rule_46`	`frontera-rich-estable`	`5.833`	`0.333`	`velocidad_constante + densidad_estable + tipo_unico + complejidad_alta + frontera_temporal + temporal_scale_stability`

world	category	mean_laws	peak_diversity	dominant signature
`rule_137`	`multiregimen-productivo`	`2.867`	`0.833`	`densidad_estable + complejidad_alta + frontera_temporal`
`rule_54`	`multiregimen-productivo`	`1.917`	`0.800`	`complejidad_alta + temporal_scale_stability`
`rule_51`	`periodicidad-global`	`4.500`	`0.333`	`periodicidad + densidad_estable + tipo_unico + complejidad_alta + temporal_scale_stability`
`rule_108`	`oscilador-local`	`2.000`	`0.167`	`periodicidad + tipo_unico`
`rule_90`	`multiregimen-escala-dependiente`	`0.500`	`0.600`	`temporal_scale_stability`

These rows show why the taxonomy cannot be reduced to a single richness score. `rule_208` and `rule_209` are maximally rich and stable; `rule_137` is less rich but highly diverse; `rule_108` is law-sparse but category-defining because it is the only local oscillator; `rule_90` has high diversity evidence but low non-empty yield because its high-scale visits become silent.

5.4 Scientific role of the atlas

The atlas is the bridge between cycle-level laws and world-level claims. A law signature describes one run. A world category describes how signatures behave across seeds, scales, and perturbations. This distinction is what makes later fragility measurements interpretable: `f_total = 0.630` in `rule_137` means something different from `f_total = 0.992` in `rule_108` because the atlas identifies different category-defining cores.

6. Fragility: `f_total`, `f_core`, `f_gap`

6.1 Protocol

Fragility is measured by exhaustive one-bit IC perturbation. For each measured world and canonical seed, every bit in the IC is flipped individually, and the resulting run is evaluated through the full pipeline (simulation, frame metrics, observers, dedup, law evaluation). The reference is the law signature of the unperturbed run.

Protocol parameters:

- **IC width**: `64` for most fragility measurements; `128` for the designed `rule_108` local-oscillator IC.
- **Perturbations per seed**: one per bit position (`64` or `128`, depending on IC width).
- **Seeds per world**: usually `3` canonical seeds, giving `192` perturbations for width-64 worlds. Designed-IC worlds such as `rule_108` use a canonical IC rather than random seeds.
- **Steps**: world-specific canonical steps (e.g., `24` for `rule_46`, `48` for `rule_137`, `96` for `rule_54`).

6.2 Metrics

Three primary fragility metrics are defined:

- **`f_total`**: fraction of perturbations that produce a different law signature from the reference (including noise-gated runs and silence).
- **`f_core`**: fraction that changes the category-defining core laws. Noise and silence count as core changes because the defining regime is lost.
- **`f_gap = f_total - f_core`**: secondary-law churn. Perturbations that change the signature without affecting the core-defining laws.

A fourth component is tracked separately:

- **`f_noise`**: fraction of perturbations that produce `analysis_status = ruido_no_analizable` (noise-gate crossing).

`f_noise` is a component of `f_total` and `f_core`; it is reported separately because it identifies a specific observer-boundary mechanism.

6.3 Core-law convention

Core laws are defined per category:

category	core laws
`frontera-rich-estable`	the full six-law frontier signature
`periodicidad-global`	`periodicidad`
`oscilador-local`	`periodicidad` and `tipo_unico`
`multiregimen-productivo`	the reference signature of that specific seed
`multiregimen-escala-dependiente`	`temporal_scale_stability`

For `multiregimen-productivo` worlds, the reference signature varies by seed. `f_core` is therefore computed per seed against that seed's reference and then averaged.

6.4 Fragility spectrum

Fase 22 completes the physical fragility spectrum for all measured non-synthetic worlds in the atlas. Synthetic controls are excluded from `f_total` and `f_core` because they are frame generators rather than evolved systems with a perturbable initial condition. The completed spectrum includes ECA worlds, Life fixtures, stable-rich frontier worlds, noise-bounded productive pockets, and the designed `rule_108` local-oscillator IC.

world	category	f_total	f_core	f_gap	f_noise	mechanism
rule_208	frontera-rich-estable	0.000	0.000	0.000	0.000	stable basin
rule_209	frontera-rich-estable	0.000	0.000	0.000	0.000	stable basin
life_block	sin-evidencia-multiregimen	0.016	0.016	0.000	0.000	stable Life fixture
rule_30	noise-bounded	0.021	0.021	0.000	0.000	productive pocket
rule_150	noise-bounded	0.023	0.023	0.000	0.000	productive pocket
rule_46	frontera-rich-estable	0.031	0.031	0.000	0.000	stable basin
life_glider	sin-evidencia-multiregimen	0.032	0.032	0.000	0.000	stable Life fixture
rule_90	multiregimen-escala-dependiente	0.172	0.000	0.172	0.000	secondary churn
rule_51	periodicidad-global	0.193	0.000	0.193	0.000	secondary churn
rule_124	multiregimen-productivo	0.224	0.083	0.141	0.000	productive switching
rule_109	multiregimen-productivo	0.307	0.307	0.000	0.000	productive switching
rule_110	multiregimen-productivo	0.323	0.198	0.125	0.000	productive switching
rule_18	multiregimen-productivo	0.349	0.135	0.214	0.000	productive switching
rule_137	multiregimen-productivo	0.630	0.312	0.318	0.000	productive switching
rule_54	multiregimen-productivo	0.714	0.677	0.037	0.375	noise-boundary
rule_108	oscilador-local	0.992	0.047	0.945	0.000	quiescent-background activation
life_blinker	sin-evidencia-multiregimen	1.000	1.000	0.000	0.000	periodic fixture disruption

The spectrum is category-aligned at the extremes: `frontera-rich-estable` occupies the low end, while `multiregimen-productivo` occupies the upper ECA range. The `life_blinker` control reaches `f_total = 1.000` because any single-cell perturbation breaks the exact Life oscillator fixture. `rule_108` remains the main ECA structural outlier: `f_total = 0.992` with only `f_core = 0.047`.

6.5 Fragility mechanisms

The atlas identifies several distinct mechanisms by which one-bit IC perturbations change law signatures:

Stable basin (`rule_208`, `rule_209`): `f_total = 0.000`. All perturbations preserve the reference signature. The basin for the six-law frontier signature is wide enough that no measured single-bit perturbation escapes it. `rule_46` is nearly identical (`f_total = 0.031`).

Stable Life fixture (`life_block`, `life_glider`): perturbations rarely change the reference signature (`f_total <= 0.032`) because the fixture remains structurally recognizable after most one-cell flips. This is fixture-level robustness, not evidence of a broad ECA basin.

Productive pocket (`rule_30`, `rule_150`): the worlds are noise-bounded in the long journal, but the non-empty pockets that survive the gate are stable under one-bit perturbation (`f_total ~= 0.02`). Their category is defined by frequent pre-law noise at scale, not by fragility of the productive signatures.

Productive basin switching (`rule_137`, and the non-noise-boundary `multiregimen-productivo` worlds): perturbations move the IC among productive law-signature regimes. The world never falls into silence or noise; `f_noise = 0.000` throughout. `rule_137` is the strongest clean case (`f_total = 0.630`), with more than 80% of perturbations switching regime in the two most fragile measured seeds.

Noise-boundary fragility (`rule_54`): perturbations cross the observer noise gate rather than moving between productive regimes. The mechanism requires complex ICs near the dedup threshold; single-bit ICs from bare backgrounds do not approach the gate (Section 8). `f_noise = 0.375` makes `rule_54` the only measured world where noise-gate crossings dominate fragility.

Periodic fixture disruption (`life_blinker`): any one-cell perturbation breaks the exact Life period-2 reference signature (`f_total = 1.000`). This is not basin switching or noise-boundary crossing; it is the brittleness of a minimal periodic fixture under full-grid perturbation.

Quiescent-background activation (`rule_108`): the canonical IC has only two active bits on a zero background. Nearly any background perturbation ignites new dynamics and changes secondary laws, producing `f_total = 0.992`. The core oscillator survives unless the perturbation lands near the motif (`f_core = 0.047`). The result is the largest `f_gap` in the atlas (`0.945`): nearly all fragility is secondary, not core.

6.6 The `f_core` / `f_gap` separation

The main result of the fragility analysis is the separation between core and secondary law changes:

- `rule_51` (`periodicidad-global`): `f_total = 0.193`, `f_core = 0.000`. Global periodicity survives all measured perturbations; only secondary laws (`densidad_estable`) toggle.
- `rule_108` (`oscilador-local`): `f_total = 0.992`, `f_core = 0.047`. The local oscillator survives nearly all perturbations; secondary laws are maximally sensitive.
- `rule_54` (`multiregimen-productivo`): `f_total = 0.714`, `f_core = 0.677`. The core productive signature changes frequently; `f_gap = 0.037` is near zero.
- `rule_137` (`multiregimen-productivo`): `f_total = 0.630`, `f_core = 0.312`. Both core and secondary transitions are common; `f_gap` is approximately equal to `f_core`.

These four cases span the space of possible (`f_core`, `f_gap`) combinations. Together they demonstrate that `f_total` alone is insufficient: two worlds can have similar total fragility with opposite core/secondary decompositions.

7. Case Studies

7.1 `rule_108` — Unique Local Oscillator

Discovery and formal profile

`rule_108` was identified during a targeted local-oscillator search (Fase 16) using minimal ICs on a quiescent background (`f(0,0,0) = 0`). The canonical IC is a pair of active cells separated by one gap (`#.#`, word `101` in binary). Under `rule_108`, this IC produces an exact period-2 local oscillator (Figure 4): the gap fills in each step (`#.# -> ###`) and then empties again, repeating indefinitely with zero drift.

Figure 4. `rule_108` period-2 oscillator motif.

```
Step t:   . . . # . # . . .
Step t+1: . . . # # # . . .
Step t+2: . . . # . # . . .   (repeats)
```

Space-time diagram to be rendered as bitmap figure. Active cells shown as `#`, quiescent background as `.`. Span <= 3, period `T = 2`.

The oscillator is stationary (center of mass fixed), bounded (`span <= 3`), and stable over 200 steps with zero drift on a uniform-zero background.

Formal profile (6 canonical seed labels, `width = 128`, `steps = 200`, IC = `pair_gap1`): `periodicidad` and `tipo_unico` are accepted in `6/6` runs. Mean `dedup_structure_count = 1.000`. The oscillator is deterministic given the canonical IC; the seed labels are retained only to keep the profile format consistent with other atlas worlds.

Algebraic derivation

The oscillator follows from three entries in the `rule_108` table:

Neighborhood	Output	Role
`010`	`1`	isolated active center stays active
`101`	`1`	gap fills in: `#.# -> ###`
`111`	`0`	center empties: `### -> #.#`

`rule_108` is left-right symmetric (`f(l,c,r) = f(r,c,l)`), which explains why the oscillator does not drift: the two flanking cells exert equal influence on the center.

Fragility: quiescent-background activation

`rule_108` has the largest fragility gap in the atlas:

Metric	Value
`f_total`	`0.992`

Metric	Value
`f_core`	`0.047`
`f_gap`	`0.945`
Core positions	`61, 62, 63, 65, 66, 67`
Pattern	`clustered`

The mechanism is geometric. The canonical IC (`pair_gap1`, 2 active bits in `width = 128`) leaves more than 120 cells at zero. A one-bit perturbation at any of those background positions activates the quiescent background: because `f(0,1,0) = 1`, an isolated `1` on a zero background immediately grows, producing new detectable structures. This changes secondary laws while leaving `periodicidad` and `tipo_unico` intact as long as the oscillator core is undisturbed. A perturbation within the core neighborhood (positions `61..63`, `65..67`) displaces or destroys the oscillator, accounting for `f_core = 0.047`.

This constitutes a third fragility mechanism, distinct from productive basin switching (`rule_137`, Section 7.3) and noise-boundary fragility (`rule_54`, Section 7.2): quiescent-background activation. The core behavior is robust, but the minimal IC makes secondary laws highly sensitive to background activation.

Uniqueness

Fase 18 ran an exhaustive search over all 128 ECA rules with quiescent backgrounds (`f(0,0,0) = 0`), testing 502 non-zero IC words per rule (all non-empty binary words of length 1..8), with `width = 128`, `steps = 200`, burn-in of 80 steps, and requiring zero drift (stationary oscillators only). Only `rule_108` produced stationary local period-2 oscillators; no other rule produced a stationary local oscillator for periods `2..16` and span `<= 32`.

A companion sweep with the stationarity requirement relaxed found a distinct family of moving local oscillators -- see Section 7.5.

The longer-IC extension strengthens the same conclusion. Extending the stationary sweep to all 7,676 non-zero IC words of length 9..12 (`982,528` rule/IC runs) produced 3,802 detections, all in `rule_108`. No new stationary oscillator rule appears beyond the length-1..8 baseline; the minimum witness is still the embedded `101` motif (`000000101` at length 9).

The family is internal to `rule_108`: 132 out of 179 candidate IC words are accepted by the production observer as `periodicidad`, with oscillator spans 3, 5, 6, 7, and 8. All confirmed oscillators have period exactly 2; no longer period was found.

7.2 `rule_54` — Noise Gate Anatomy

`rule_54` is the clearest example of noise-boundary fragility: perturbations do not merely move the run to another productive signature, but can push the observer output across the deduplicated structure gate. The production gate is:

```
dedup_structure_count > 40 -> ruido_no_analizable
```

Fase 13: anatomy of the gate

Fase 13 measured three productive `rule_54` ICs at `steps = 96` and perturbed each by all 64 one-bit flips. The reference deduplicated counts were close to the gate:

seed	reference dedup	noisy flips / 64
`20260638`	`32`	`14`
`20260640`	`33`	`18`
`20260642`	`39`	`40`

Across the three seeds, `72/192` flips crossed into `ruido_no_analizable` (`f_noise = 0.375`). Every noisy flip crossed for the same reason: `dedup_structure_count > 40`. No alternative noise mechanism was observed.

The sensitive positions formed a clustered, multi-hot pattern rather than a single contiguous block. Bins near the periodic boundary (`0..7` and `56..63`) were repeatedly implicated, and bit 5 was the only bit whose flip crossed the gate in all three measured seeds.

Fase 19: controlled single-bit negative case

Fase 19 tested whether bit 5 was a special absolute coordinate of `rule_54`. It replaced the complex ICs with controlled single-bit ICs: for each `k = 0..63`, the initial state contained only one active bit at position `k`.

The result separates CA physics from the observer pipeline:

- The ECA frames are translation-invariant after shift normalization.
- The observer/dedup counts are not translation-equivariant for this wide-spreading pattern: `dedup_structure_count` ranges from `15` to `24` across positions.
- The law signature is identical for all 64 positions: `temporal_scale_stability`.
- Every single-bit IC remains far below the gate (`dedup <= 24 < 40`).

Therefore, bit 5 is not a privileged coordinate of `rule_54`. The Fase 13 signal arises from the interaction between complex IC geometry and the observer/gate pipeline. Complex ICs close to the threshold can be pushed across it by local flips; a single active cell cannot.

Mechanism

`rule_54` has high total and core fragility (`f_total = 0.714`, `f_core = 0.677`), but its mechanism differs from `rule_137`. In `rule_137`, perturbations tend to move between productive regimes. In `rule_54`, a large fraction of perturbations cross an analysis boundary: the run becomes too fragmented for the current observer/dedup gate.

This makes `rule_54` a methodological case study as much as a dynamical one. It shows that the atlas can identify worlds whose measured fragility is dominated by proximity to an observer threshold. It also motivates the caveat that absolute structure counts should not be treated as symmetry-invariant physical observables without equivariance checks.

7.3 `rule_137` — Productive Basin Switching

`rule_137` is the primary example of productive basin switching: one-bit IC perturbations change the law signature without ever crossing the noise gate or reaching silence. All fragility is productive (`f_noise = 0.000`, `f_silence = 0.000`), making it the cleanest case in the atlas for inter-basin transitions.

Fragility profile

Three canonical seeds at `steps = 48`, `width = 64`:

seed	reference signature	f_total
`20260633`	`complejidad_alta + frontera_temporal`	`0.812`
`20260635`	`complejidad_alta + densidad_estable + frontera_temporal`	`0.219`
`20260673`	`complejidad_alta + densidad_estable + frontera_temporal + temporal_scale_stability + tipo_unico + velocidad_constante`	`0.859`

Aggregate: `f_total = 0.630`, `f_core = 0.312`, `f_gap = 0.318`.

The per-seed range (`0.219..0.859`) is the widest in the atlas. Even the least fragile measured seed has `f_total > 0.2`. The most fragile seeds (20260633 and 20260673) flip on more than 80% of one-bit perturbations.

Mechanism

`f_noise = 0.000`: no perturbed IC crosses the deduplicated structure gate. The world remains analyzable throughout. The fragility is a property of productive basin geography, not proximity to an observer threshold.

The pattern is `dispersed`: sensitive positions are distributed across the IC width rather than concentrated near a motif. This is consistent with a world that has many narrow productive basins whose boundaries intersect throughout the IC space.

`peak_diversity = 0.833` — the highest in the atlas. The canonical seeds themselves already visit multiple distinct productive regimes. The fragility measurement extends this: not just that the world can reach different signatures under different seeds, but that a single-bit perturbation to any one canonical IC is enough to move between regimes.

f_core and f_gap interpretation

`f_core = 0.312` reflects genuine regime switching: flips that remove or change laws defining the canonical signature. `f_gap = 0.318` reflects secondary-law churn: the core productive signature survives, but laws on the signature boundary (such as `densidad_estable` or `tipo_unico`) toggle.

The two components are roughly equal (0.312 vs 0.318), meaning `rule_137` sits in a region where both core-regime transitions and secondary-law transitions are common. This is structurally different from `rule_108` (`f_gap = 0.945`, where the core oscillator is robust and secondary laws dominate) and from `rule_54` (`f_gap = 0.037`, where the core productive signature changes but almost no fragility is secondary).

Contrast with `rule_54` and `rule_108`

`rule_54` and `rule_137` both have high `f_total` (0.714 vs 0.630), but the mechanisms are opposite: `rule_54` fragility is dominated by noise-gate crossings (`f_noise = 0.375`), while `rule_137` fragility is entirely productive. A perturbed `rule_137` IC stays analyzable and law-rich; it is simply in a different productive regime.

`rule_108` contrasts from the opposite direction: its `f_core = 0.047` shows that the defining behavior (the local oscillator) is nearly indestructible, while `rule_137`'s `f_core = 0.312` shows that its defining signatures change under nearly a third of all one-bit flips.

7.4 `rule_46`, `rule_208`, `rule_209` — Stable-Rich Frontier

These three worlds define the `frontera-rich-estable` category: low signature diversity, near-maximal law richness, and very low fragility. They are the counterexample that revised the early atlas interpretation of `frontera_temporal`.

The first 15-world atlas made `frontera_temporal` look rare: it appeared only as a minority law in class-4 multi-regime worlds such as `rule_137`, `rule_110`, and `rule_54`. Fase 11 showed that this was a sampling artifact. A sweep over all 256 ECA rules (`seeds = 20260523..20260525`, `W = 64`, `T = 24`) found 38 rules where `frontera_temporal` activates in at least two of three seeds, and 17 rules where it activates in all three.

The top rules by law richness were `rule_46`, `rule_208`, and `rule_209`. Formal six-seed profiles (`20260523..20260528`, `W = 64`, `T = 24`) placed all three in a new category:

world	mean laws	peak diversity	category
`rule_46`	5.833	0.333	`frontera-rich-estable`
`rule_208`	6.000	0.167	`frontera-rich-estable`
`rule_209`	6.000	0.167	`frontera-rich-estable`

The dominant signature is the same six-law set:

```
velocidad_constante
densidad_estable
tipo_unico
complejidad_alta
frontera_temporal
temporal_scale_stability
```

Only `periodicidad` is absent. This makes the family nearly maximal under the current seven-law system without relying on multi-regime exploration.

Stable richness rather than multi-regime diversity

The category is defined by the conjunction of high richness and low diversity. `rule_137` is rich because it moves among several productive law signatures. The frontier-rich worlds are rich because the same high-law signature appears reliably across seeds.

This distinction matters operationally. A policy that only looks for signature diversity would miss these worlds, even though they produce more accepted laws per visit than any multi-regime world in the atlas. The Fase 11 taxonomy update therefore adds:

```
frontera-rich-estable := mean_laws >= 4.0 and peak_diversity <= 0.5
```

evaluated after noise-bounded and multi-regime cases.

Complement symmetry and independent convergence

`rule_46` and `rule_209` are a complement pair (`46 = 255 - 209`): exchanging zeros and ones maps one into the other. Their shared profile is therefore one physical phenomenon seen through global bit inversion.

`rule_208` is more surprising. Its complement is `rule_47`, not `rule_46` or `rule_209`, yet it reaches the same maximum-richness profile. This suggests that the `frontera-rich-estable` regime is not a single isolated symmetry orbit; at least two distinct ECA regions converge to the same six-law frontier.

Fragility

Fase 12 measured one-bit fragility for the three worlds using the same protocol as `rule_137` and `rule_54`:

world	f_total	f_core	f_noise
`rule_46`	0.031	0.031	0.000
`rule_208`	0.000	0.000	0.000
`rule_209`	0.000	0.000	0.000

The result is the opposite end of the fragility spectrum from `rule_137`. Where `rule_137` has many narrow productive basins (`f_total = 0.630`), `rule_208` and `rule_209` have measured basins so wide that no single-bit flip changes the law signature. `rule_46` is only slightly fragile: two of 192 single-bit perturbations change signature across the three measured seeds.

This confirms that high law richness does not imply high fragility. Richness can arise either from many neighboring productive regimes (`rule_137`) or from a single broad, stable regime (`rule_46/208/209`).

Scientific revision

The correct conclusion is not that `frontera_temporal` is intrinsically rare. It is rare in the original discovery atlas because the original world sequence under-sampled stable high-richness boundary worlds. In the full ECA sweep, `frontera_temporal` is a robust marker of the `frontera-rich-estable` family.

7.5 Moving Oscillator Family -- Minimal Period-2 Gliders

A companion sweep to Fase 18 searched all 128 quiescent ECA rules (`f(0,0,0) = 0`) for oscillators that translate at constant velocity. The detection protocol matched Fase 18 except: the zero-drift requirement was replaced by a requirement of constant non-zero drift confirmed over three consecutive periods; `width` was extended to `256` and `steps` to `300` to give moving patterns room to travel. The 502 non-zero IC words of length 1..8 were tested per rule (64,256 total runs, 312 s).

Eight rules produce moving oscillators: `rule_6`, `rule_20`, `rule_38`, `rule_52`, `rule_134`, `rule_148`, `rule_166`, `rule_180`. All share the same minimal glider pattern:

Step	Active shape (normalized offsets)
`t`	`[0]` -- single active cell
`t+1`	`[0, 1]` -- two adjacent active cells
`t+2`	`[0]` displaced by `+/-2`

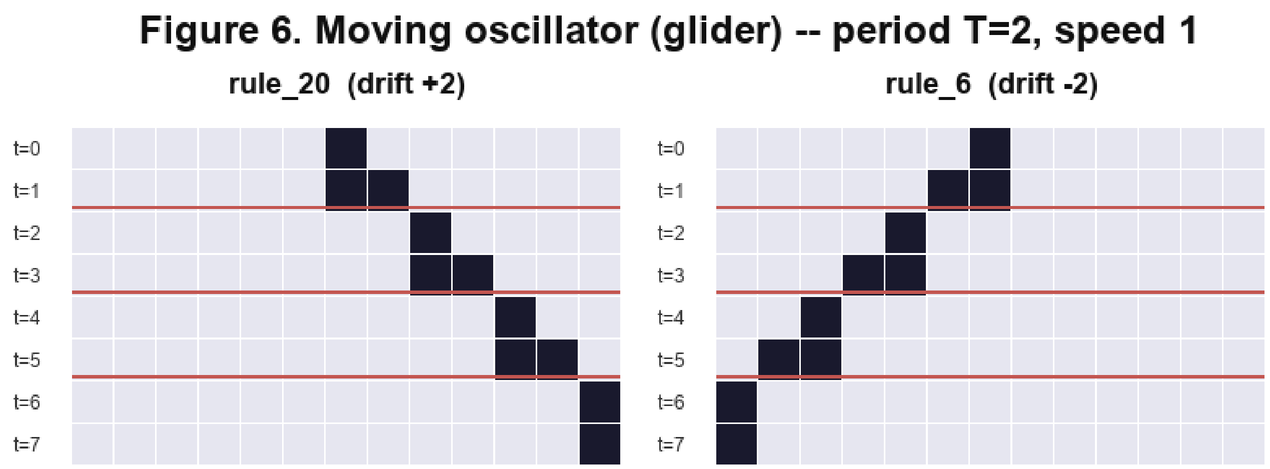
The oscillator alternates between one and two active cells while traveling at speed 1 cell per step -- the maximum velocity for a radius-1 ECA. Mean active span per period is 0.5 (alternating span 0 and span 1). All eight rules are confirmed `edge_touch = False` within `width = 256`.

Figure 6. Moving oscillator (glider) -- period T=2, speed 1.

```
rule_20 (drift +2)           rule_6 (drift -2)

t=0:  . . . 1 . . . . .    . . . . . 1 . . .
t=1:  . . . 1 1 . . . .    . . . . 1 1 . . .
t=2:  . . . . . 1 . . .    . . . 1 . . . . .    <- period boundary
t=3:  . . . . . 1 1 . .    . . 1 1 . . . . .
t=4:  . . . . . . . 1 .    . 1 . . . . . . .    <- period boundary
```

Active cells shown as `1`, quiescent background as `.`. Each period advances the pattern two positions in the travel direction.



Left: rule_20 (right-moving, drift +2 per period). Right: rule_6 (left-moving, drift -2 per period). Active cells dark; quiescent background light. Rows are time steps t=0..7. The oscillator alternates [0] <-> [0,1] while translating one cell per step.

Structure: four mirror pairs

The eight rules form four left-right symmetric pairs:

Left-moving	Right-moving	`b5`	`b7`
`rule_6`	`rule_20`	0	0
`rule_38`	`rule_52`	1	0
`rule_134`	`rule_148`	0	1
`rule_166`	`rule_180`	1	1

Bits `b5` (neighborhood `101`) and `b7` (neighborhood `111`) vary across the family but do not participate in the glider cycle: neither neighborhood occurs during quiescent travel with a two-cell active pattern. The eight rules are therefore a single physical family parametrized by two inactive bit choices and the left-right direction.

Contrast with `rule_108`

Property	`rule_108`	Moving family
Drift per period	0	+/-2
Period `T`	2	2
Mean active span	~2	0.5
Speed	0 (stationary)	1 (maximum)
Rules in scope	1	8 (4 mirror pairs)

The two families do not overlap. `rule_108` does not appear among the eight moving rules, and none of the eight moving rules produce stationary local oscillators under Fase 18. Together, the two sweeps partition the quiescent local oscillator landscape into a unique stationary oscillator (`rule_108`) and a unique minimal glider family (8 rules, 4 mirror pairs).

Uniqueness within the protocol

No IC word of length 1..8 produced a moving oscillator with `T > 2` or `|drift| != 2` in any quiescent rule. The minimal glider `[0] <-> [0, 1]` at period 2 and maximum speed is the only moving local oscillator family under this protocol.

The longer-IC extension reaches the same result. Testing all 7,676 non-zero IC words of length 9..12 over all 128 quiescent rules (`982,528` rule/IC runs) produced 2,059 moving detections, all within the same eight-rule family. No new moving-oscillator rule appears. The minimal witnesses are still the old glider seeds embedded in longer words (`000000001` for right movers, `000000010` for left movers). The sweep also filtered 9,822 period-1 moving particle aliases across 32 rules; these are moving particles, not internal period-2 oscillators. Extensions to IC words longer than 12 remain open (Section 10.2). The non-zero background extension is reported in Section 7.6.

7.6 Periodic-background oscillator sweep

The two sweeps above restrict the background to quiescent zero cells. A third sweep tests whether replacing the background with a non-zero periodic tiling changes the oscillator landscape.

Protocol. All 256 ECA rules are tested against 15 unique non-zero periodic backgrounds with template lengths 1, 2, and 4. Each rule/background pair is tested against 502 non-zero IC words of length 1..8 centered in a width-256 grid for 300 steps with 80 burn-in. The detector identifies exact recurrence of the localized difference between the perturbed run and the unperturbed background orbit; global background periodicity alone is not counted as a local oscillator. Total: 1,927,680 rule/background/IC runs, 122,253 candidate detections.

Result. Thirty rules produce stationary local oscillators under at least one non-zero periodic background; 36 rules produce moving oscillators. Of the 30 stationary rules, 29 are new relative to the zero-background baseline. Of the 36 moving rules, 28 are new.

The following phenomena appear under non-zero backgrounds but not under the quiescent zero background:

- **Period-4 stationary oscillators.** `rule_54` and `rule_147` produce stationary period-4 oscillators under `0001` background.
- **Period-4 moving oscillator.** `rule_180` produces a T=4 glider with drift +4 under `0001` background (shapes `[0] -> [0,1] -> [0,2,3] -> [0]`), distinct from its T=2 speed-1 glider under quiescent background.
- **Speed-0.5 gliders.** Multiple rules (including `rule_3`, `rule_17`, `rule_27`, `rule_35`, `rule_39`) produce T=2 gliders with drift +/-1 under non-zero backgrounds, corresponding to 0.5 cells per step. Under quiescent zero background the only observed glider speed was 1 cell/step.
- **`rule_108` under all-one background.** `rule_108` appears in the stationary list with background `1` and the same motif `### / #.#` as the quiescent result, confirming that the rule_108 oscillator is intrinsic to the rule table.

Separation of regimes. The zero-background uniqueness claims (Sections 7.1 and 7.5) are not contradicted: they describe the quiescent regime. Under zero background, the quiescent local oscillator space is sparse (1 stationary rule, 8 moving rules); under non-zero periodic backgrounds, the landscape expands substantially (30 stationary, 36 moving) and includes period and speed classes absent from the quiescent regime. The two regimes should not be merged.

7.7 Period-8 background oscillator sweep (Fase 24)

A fourth oscillator sweep extended the periodic-background protocol to length-8 primitive binary backgrounds, testing three questions: do longer background periods introduce new oscillator rules, new period classes T>4, or glider speeds outside {0, 0.5, 1}?

Protocol. All 256 ECA rules are tested against 30 primitive binary necklaces of length 8. A length-8 binary string is primitive if its minimal period is exactly 8; the necklace representative is the lexicographically smallest member of its rotation class. The count of 30 follows from Mobius inversion: `(2^8 - 2^4) / 8 = 30`. Each rule/background pair is tested against 502 non-zero IC words of length

1..8 in a width-256 grid, 300 steps, 80 burn-in. The differential detector and period search window (2..16, span <= 32) are unchanged from Section 7.6. Total: 3,855,360 runs.

Result. The sweep produces 323,872 candidate detections after filtering 95,121 period-1 aliases. New stationary rules beyond the length-1/2/4 baseline are `rule_62`, `rule_118`, `rule_131`, and `rule_145` (4 rules, all T=3). New moving rules are `rule_7`, `rule_9`, `rule_21`, `rule_25`, `rule_31`, `rule_45`, `rule_61`, `rule_65`, `rule_67`, `rule_75`, `rule_87`, `rule_88`, `rule_89`, `rule_101`, `rule_103`, `rule_111`, `rule_125`, `rule_173`, and `rule_229` (19 rules).

New period classes. Periods T=6, 8, 10, 12, and 15 appear for the first time; under backgrounds of length 1, 2, and 4 the maximum observed period was T=4. T=15 is a non-trivial period not divisible by the background length (8). Fase 24 reports it as an emergent period; Sections 7.8 and 7.9 subsequently identify its rule family and establish its five-state mechanism under `F^3`.

New glider speed. Speed 2/3 cell/step (drift +/-2, T=3) is observed for the first time; the prior speed set was {0, 0.5, 1}. Representative cases are `rule_9` (drift -2, T=3, background `00001001`) and `rule_65` (drift +2, T=3, background `00000001`). Two further rules with the same speed signature, `rule_111` and `rule_125`, appear in the candidate table. Direct rule-table reflection, `g(l,c,r) = f(r,c,l)`, confirms two exact left-right mirror pairs: `rule_9 <-> rule_65` and `rule_111 <-> rule_125`. The paired rules carry opposite drift signs with the same T=3 speed magnitude, giving the speed-2/3 family the same left/right mirror structure as the speed-1 family in Section 7.5.

Phase dependence. A rotation sub-test applied all 8 rotations of the canonical background to 10 sampled rules while holding the IC fixed. None of the 10 samples is active in all eight rotations after circular-geometry correction. `rule_62` and `rule_118` activate in 7 of 8 background rotations; moving cases range from 6/8 (`rule_9`, `rule_65`) to 2/8 (`rule_111`, `rule_45`). The IC therefore requires a particular alignment with the background to nucleate the oscillator.

Co-translation test (Fase 25). A strict test co-translates both background and IC through `k=0..7` for the same 10 cases. The background and XOR perturbation orbits are exact translations in 80/80 runs. The original `linear_shape` preprocessing recovers only 58/80 signatures: 22 moving runs cross positions 255/0, causing a false linear span and rejection. Circular shape canonicalization (largest-gap cut plus continuous position unwrapping) recovers 80/80 signatures and all 10/10 cases. Thus the physics and recurrence detector are co-translation equivariant once cyclic geometry is represented correctly. The remaining fixed-IC phase dependence is physical, although the linear observer overstated its severity for moving rules.

Summary. All three Fase-24 questions are answered affirmatively: period-8 backgrounds introduce 4 new stationary rules and 19 new moving rules, expand the period set to include T=6, 8, 10, 12, and 15, and introduce speed 2/3 cell/step as a new rational class. The zero-background uniqueness claims of Sections 7.1 and 7.5 are unaffected.

7.8 Anatomy of the T=15 family (Fase 26)

The longest period observed in Fase 24 was analyzed separately to determine whether it represented a single accidental witness or a coherent family. All 221 `T=15` detections are stationary and occur in only two rules: `rule_73` (123 detections) and `rule_109` (98). Both rules are left-right symmetric, and black/white conjugation maps each rule exactly to the other. The detections cover 14 primitive length-8 backgrounds, 20 rule/background pairs, and 25 temporal motifs up to cycle phase. The minimum witness is `rule_109` on background `00011001` with the two-cell IC word `01`.

Temporal locking. Every participating unperturbed background enters a temporal orbit of period `T_bg=3` (with transient length 0..2). The localized perturbation has fundamental period `T_local=15`, giving the same locking ratio `T_local/T_bg=5` in all 20 rule/background pairs. Thus `T=15` is not inherited directly from the spatial background length 8. Long-horizon reruns of one minimal witness per pair preserve exact recurrence through step 900 in 20/20 cases; the detector scans upward from period 1 and therefore excludes smaller fundamental periods.

Basin width. Persistence in time does not imply robustness to initialization. Holding the IC fixed while rotating the background preserves `T=15` in only 23/160 runs. One-bit mutations of the 20 minimal IC witnesses preserve it in 4/134 runs. Most failed perturbations settle into shorter localized periods `T=3` or `T=6`; a small number produce `T=12` or no localized period in the search window. The family is therefore temporally exact but basin-narrow.

These results establish a persistent background-locked family rather than a single numerical coincidence. Fase 26 measures but does not explain the five-to-one locking ratio; Fase 27 addresses its finite-state mechanism next.

7.9 Five-state locking mechanism (Fase 27)

Fase 26 measured `T_local/T_bg=5` across all 20 minimal `T=15` representatives but left the origin of that ratio open. Fase 27 closes the computational half of the question by tracking the localized XOR defect `D(t) = X(t) XOR B(t)` once per background period.

Protocol. After burn-in, the defect is sampled at `t = 81 + k*T_bg` for `k=0..20`, covering four complete candidate five-cycles for each of the 20 minimal `(rule, background, IC)` witnesses from Fase 26. Acceptance requires eight simultaneous checks: the background returns to the exact same phase at every sample; the first five defect states are mutually distinct; the fifth transition closes the cycle; no shorter cycle under `F^3` explains the sequence; four consecutive cycles repeat in both canonical and raw-position encodings; transitions remain deterministic across cycles; and the oscillator is stationary over every local period.

Results. All eight checks are satisfied for 20/20 representatives. Every `T=15` oscillator in the family is stationary (`drift=0`). The defect cycles through exactly five distinct states under `F^3`, the three-step evolution operator, and five applications of `F^3` are necessary and sufficient to restore the complete background-plus-defect configuration. The same result holds over four consecutive cycles in both relative-shape and absolute-position encodings.

Interpretation. The locking ratio `T_local/T_bg=5` is the cycle length of the defect under `F^3`, not a resonance with the spatial background length 8. The result is a computational state-cycle derivation: it establishes the finite-state mechanism without reducing the five nodes to a closed-form symbolic identity over the rule-table algebra of `rule_73/rule_109`. That symbolic derivation remains open.

7.10 Induced defect rule and exact conjugation (Fase 28)

Fase 28 examines the local algebra inside the five-state cycle. A localized XOR defect does not evolve under the original ECA rule `f` in isolation. If `b` is a three-bit background neighborhood and `d` its XOR-defect neighborhood, the exact induced update is

`delta_f(b,d) = f(b XOR d) XOR f(b)`.

The analysis profiles this induced rule over all 300 microsteps contained in the 100 `F^3` edges from the 20 minimal representatives.

Analytical conjugation. Let `C` denote bitwise complementation. Since `rule_109` is the black/white conjugate of `rule_73`, their global maps satisfy `F_109(C(X)) = C(F_73(X))`. If both the full state and background are complemented, induction gives `X_109(t)=C(X_73(t))` and `B_109(t)=C(B_73(t))` for every `t`. Therefore:

`D_109(t) = C(X_73(t)) XOR C(B_73(t)) = D_73(t)`.

At local level, the equivalent identity is `delta_109(C(b),d) = delta_73(b,d)`. Thus the defect orbit is invariant, not complemented, under the simultaneous black/white conjugation. This statement is analytical; exhaustive checks over all 64 local `(b,d)` combinations and 10/10 complemented orbit pairs serve as implementation validation.

Sparse-support hypothesis rejected. Every one of the 100 `F^3` macro-transitions uses all eight ordinary truth-table entries in its causal defect cone. No single induced `(b,d)` key appears in every macro-transition. The five-cycle therefore cannot be explained by one fixed sparse subset of local entries.

The negative result still exposes phase structure. The sizes of the induced-key intersections for transitions `S0->S1` through `S4->S0` are `[9,6,7,5,4]` for `rule_73` and `[11,11,10,9,8]` for `rule_109`. These signatures are rule-specific rather than universal. A closed-form derivation must therefore encode spatial phase or a higher-order block state, rather than truth-table entry presence alone.

7.11 Block-locality limits and background-indexed shape families (Fase 29–30)

Section 7.10 concludes that a closed-form derivation of the five-cycle must encode spatial phase rather than truth-table entry presence alone. Fase 29 and Fase 30 test two complementary refinements of that conclusion.

Defect-shape locality (Fase 29). For each of the 20 minimal representatives and each of the five cycle phases, the canonical XOR-defect shape is computed at sample times `t = 81, 84, 87, 90, 93`. If the T=15 cycle were a pure defect-only dynamic, every background would produce the same canonical shape in each phase. Instead, `rule_73` produces 8–9 distinct shapes per phase across its 10 backgrounds, and `rule_109` produces 8. Extending the comparison window outward to fixed local blocks (active defect span plus up to three padding cells on each side) does not help: no nontrivial block signature is shared across all backgrounds in any phase. The `W=0` result in the block-signature scan shows that the active defect span itself already carries background-dependent context — no additional padding is needed before the discrimination occurs. Only the trivial token `d000->0` (background cells evolving to background under all three microsteps) is universal.

Shape families (Fase 30). Although the defect shapes are background-dependent, they are not unstructured. Treating two five-state cycles as equivalent when one is a cyclic phase rotation of the other, the 20 representatives decompose into 7 distinct families for `rule_73` and 8 for `rule_109`, yielding 13 global families. The largest family has 3 members. Two families are shared across the conjugate rules: one of size 3 (two `rule_73` backgrounds and one `rule_109` background) and one of size 2 (background word `00110101` producing phase-rotationally equivalent defect cycles under both rules). This second coincidence is not implied by the analytical conjugation of Fase 28 — the bitwise complement of `00110101` is `11001010`, a distinct word — so it is an independent structural coincidence. Scalar background descriptors (`active_count`, `transition_count`, `active_transition_pair`) do not determine the family. The canonical temporal orbit of the background under the same rule is exact in this representative set, but this is largely a restatement of full background identity rather than a compact symbolic law.

Interpretation. Fase 29 and Fase 30 together bracket the derivation problem: a correct symbolic account must use the full temporal background orbit (or an equivalent compact encoding), and maps that orbit to one of a finite set of defect-cycle shapes plus a phase offset. A derivation predicting a single universal five-state cycle is falsified by Fase 29; a derivation predicting arbitrary unclustered shape variation is refuted by Fase 30.

7.12 Compact state variable for the T=15 family (Fase 31–32)

Fase 30 reduced the T=15 family to 13 shape families but required the full temporal background orbit as the discriminating descriptor. Fase 31 and Fase 32 search for a shorter description.

Descriptor search (Fase 31). A compact global background descriptor does not exist among the candidates tested (length-2..4 circular subpattern counts, parity, run lengths, and orbit prefixes up to 24 bits). Globally, only `orbit_prefix_24` determines the family, which is substantially a restatement of full background identity. Conditioned on the ECA rule, the shortest non-orbit candidate is the circular multiset of length-4 background subwords (`subpatterns_len4`): it separates all 10 backgrounds per rule into distinct buckets with zero ambiguity. A global decision tree over all features achieves only 0.700 training accuracy at depth 4, confirming that no shallow feature combination separates all 13 families simultaneously.

Rotation generalization (Fase 32). The circular subpattern multiset is invariant under background rotation. Fase 32 tests whether the predicted family is preserved across all seven non-trivial rotations of each of the 20 representative backgrounds, under two modes: `fixed_ic` (background rotated, IC position fixed) and `cotranslated_ic` (background rotated, IC shifted by the same amount to preserve local alignment).

Under `fixed_ic`, only 3 of 140 rotations produce T=15 at all, and 1 matches the predicted family. Under `cotranslated_ic`, all 140 of 140 rotations produce T=15 and all 140 match the predicted family. The co-translation result is not evaluated on new backgrounds but on rotational variants of the known 20 representatives; it validates that the descriptor is rotation-equivariant within the confirmed family set, not that it predicts previously unseen data.

Compact state variable. The complete minimal description consistent with all observations is the triple `(rule, subpatterns_len4, IC/background alignment)`. The rule identity determines which of the two conjugate families applies; the subpattern multiset identifies the shape-family class within that rule; and the IC/background alignment selects the phase within the five-state cycle. A derivation operating on the background word alone without IC alignment is falsified by the `fixed_ic` mode (3/140 detections). Any derivation that correctly maps this triple to a defect cycle shape and phase offset is a complete symbolic account of the T=15 family.

8. Observer Artifacts and Pipeline Equivariance

The ZUSE pipeline contains two classes of observer artifact that the atlas identifies and characterizes. Framing these artifacts as results — not just implementation limitations — is important: they define the boundary between what the system measures reliably and what it does not.

8.1 Mirror asymmetry in `tipo_unico` (Fase 6b)

`rule_110` and `rule_124` are left-right mirrors of each other: the rule table for `rule_124` is obtained by reflecting every neighborhood `f(l,c,r) -> f(r,c,l)` in the `rule_110` table. Under periodic boundary conditions, the two rules produce physically equivalent dynamics up to spatial reflection.

Despite this equivalence, `tipo_unico` can fire asymmetrically: it may be accepted for one orientation and rejected for the other, depending on which structure types the heuristic observers label from the specific frame sequence. Since `tipo_unico` counts whether exactly one structure type appears, its value depends on the labeling convention of the observers, not only on the CA dynamics.

`tipo_unico` is retained in the atlas for its exploratory value: it reliably distinguishes runs with homogeneous structure populations from runs with mixed populations. It should not be used as evidence of physical left-right asymmetry.

8.2 Translation non-equivariance of the dedup pipeline (Fase 19)

ECA with periodic boundary conditions is translation-invariant: shifting the initial condition by any number of cells produces the same dynamics up to a spatial shift. A pipeline that correctly identifies physical structures should therefore return the same structure count for all translations of the same IC.

Fase 19 tested this directly: 64 single-bit ICs for `rule_54`, one active bit at each position `k = 0..63`, with `width = 64` and `steps = 96`. The ECA frames were confirmed translation-invariant (frame identity after shift normalization: `True`). The observer/dedup pipeline was not:

metric	range across 64 positions
`dedup_structure_count`	`15..24` (29 distinct result classes)
`raw_structure_count`	`45..72`
law signature	`temporal_scale_stability` (all 64 identical)
`analysis_status`	`ok` (all 64)

metric	value
ICs tested	64 single-bit positions (`k = 0..63`)
ECA translation-invariant	True

metric	value
Unique result classes (observer)	29
dedup range observed	15-24
Noise gate threshold	> 40
Closest single-bit IC to gate	`k = 55`, `dedup = 24`
Complex-IC reference (Fase 13)	`dedup = 32..39`
Law signature (all 64 positions)	`temporal_scale_stability`
Conclusion	noise-gate crossing requires complex IC geometry; single-bit ICs stay at least 16 below gate

The mechanism is a boundary interaction: `rule_54` produces wide-spreading patterns that cross the periodic frame boundary. The dedup algorithm's handling of cyclic-span structures varies depending on the absolute IC position relative to where structure boundaries fall on the lattice. The result is a position-dependent count that is not a translation-equivariant physical observable.

Fase 25 isolates a second instance of the same geometric failure mode in the local-oscillator detector. Co-translated background and XOR perturbation orbits are identical in 80/80 runs, but `linear_shape` loses 22 moving signatures when the localized difference straddles positions 255 and 0. Circular canonicalization restores 80/80 signatures. This directly attributes the non-equivariance to linear treatment of a periodic lattice.

8.3 Implications for the atlas

Both artifacts are bounded in their effect:

- `tipo_unico` asymmetry is a labeling artifact, not a count artifact. It affects which laws are accepted, but only for runs where the structure population is near the one-type boundary. Runs with clearly homogeneous or clearly mixed populations are unaffected.
- Dedup non-equivariance affects absolute structure counts but not law signatures. In Fase 19, all 64 translated ICs produce identical law signatures despite varying dedup counts. The noise gate (`dedup > 40`) is never approached by single-bit ICs, and law evaluation depends on count magnitude only through the gate.

The atlas therefore relies on law signatures as the primary evidence unit. Absolute dedup counts appear in world profiles as context and should be interpreted with the translation-equivariance caveat. Future work on symmetry-invariant observers would remove both artifacts.

9. Limitations

9.1 Fixed protocol parameters

The atlas is valid for the parameter regime used: `width = 64` (formal profiles), `width = 128` (rule_108 oscillator), `steps` roughly `24..200`, and the IC protocols defined per world. The two calibrated thresholds — `frontera_temporal` upper bound `0.4352` and `temporal_scale_stability` threshold `19.03` — were fit on data from this regime. Applying the atlas to significantly different widths or step counts requires recalibration. This is not a flaw in the methodology; it is the expected scope of an empirically grounded atlas.

9.2 Heuristic observers

The observer stack uses geometric heuristics to label structures as `glider`, `bloque`, or `oscilador`. These heuristics are not derived from first principles and are not provably complete or sound for arbitrary ECA dynamics. As shown in Section 8, they are not translation-equivariant for wide-spreading patterns and are not mirror-invariant for `tipo_unico`. The atlas is built on law signatures, which are more robust than absolute observer counts, but the underlying observers remain heuristic. Replacing them with symmetry-invariant observers would be a meaningful improvement.

9.3 Bounded local oscillator protocol

The uniqueness claim for `rule_108` holds under a specific protocol: quiescent zero background, stationary exact periodicity (no drift), IC words of binary length 1..12 (502 words of length 1..8 plus 7,676 words of length 9..12), and period detection window 2..16 with local span <= 32. IC words longer than 12 or longer detection periods are outside the current stationary protocol. A separate periodic-background sweep (Section 7.6) confirms that non-zero backgrounds substantially change the oscillator landscape; those results define a different regime and do not apply to the zero-background uniqueness claim. Moving oscillators under quiescent zero background were searched in a companion sweep -- Section 7.5 reports the result. The claim is therefore: no other quiescent ECA rule produces a stationary local-period oscillator under this protocol. It is not a claim about ECA oscillators in general.

The period-8 background sweep (Section 7.7) is also bounded by background phase. Fase 25 completed the strict co-translation test: the physical orbits are exact translations in 80/80 runs, while the original linear observer recovers 58/80 signatures because of cyclic boundary crossings. Circular canonicalization recovers 80/80. Fixed-IC phase dependence remains in all 10 sampled cases after correction, so it is physical alignment sensitivity rather than pure observer non-equivariance.

Fase 26 strengthens the longest-period result without converting it into a general theorem. All 20 minimal `T=15` rule/background representatives persist through step 900 and share a background temporal period of 3, but only 23/160 background phases and 4/134 one-bit IC mutations retain `T=15`. The result is therefore robust in time and narrow in basin under the tested protocol.

Fase 27 establishes the state-cycle mechanism of the 5:1 locking computationally. After burn-in, the localized defect `D(t) = X(t) XOR B(t)` cycles through exactly five distinct states under `F^3` across all 20 minimal representatives (Section 7.9). The 5:1 ratio is therefore the cycle length of the defect under the three-step operator, not a spatial resonance. The symbolic derivation of why this five-cycle appears in `rule_73/rule_109` but not in other rules over `T_bg=3` backgrounds remains open.

Fase 28 proves the black/white conjugation of the defect dynamics analytically and verifies it exhaustively at local and orbit levels (Section 7.10). It also rules out one tempting simplification: no fixed sparse subset of induced local transitions is shared by all 100 `F^3` edges. The remaining symbolic problem requires phase-aware or higher-order block variables.

Fase 29 and Fase 30 sharpen that boundary. The T=15 cycle is not a pure defect-only dynamic: canonical defect shapes are background-specific already at `W=0`, and no nontrivial fixed local block signature is shared across all backgrounds. But the variation is not arbitrary. The 20 representatives collapse into 13 phase-rotated defect-cycle families, including two families shared across `rule_73` and `rule_109`. The open problem is therefore not simply "find the local rule"; it is to map the temporal background orbit and IC alignment to a finite shape family and phase offset.

Fase 31 and Fase 32 provide the current compact description of that map. No tested background-only descriptor shorter than the full temporal orbit predicts the 13 families globally. However, conditioned on rule identity, the circular multiset of length-4 background subwords separates the families, and rotational variants preserve the predicted family in 140/140 co-translated runs. The same rotations fail almost completely with fixed IC placement (3/140 detections), so IC/background alignment is an essential physical variable, not a bookkeeping detail.

9.4 Empirical atlas, not axiomatic classification

The world categories are induced from observed law signatures across a finite number of seeds and step counts. A world classified as `multiregimen-productivo` on 6..15 visits could exhibit different behavior at larger scale, with different IC distributions, or under longer runs. The categories are stable empirical summaries, not theorems. `rule_90` is a clear example: it is classified as `multiregimen-escala-dependiente` because high-scale visits become silent under the current protocol, but the underlying XOR dynamics have algebraic structure that the current seven laws do not capture.

Fase 20 gives the same warning for `frontera_temporal`: 24 additional rules were rich in `frontera_temporal` at sweep scale, but the top four failed long-journal validation. Category assignment is therefore protocol-scale dependent; short-scale richness is candidate evidence, not atlas-grade classification.

Fase 21 gives an analogous warning for `periodicidad`. With random ICs, production `periodicidad` is rare in ECA; with explicitly periodic IC families, it appears in `207/256` rules. The law is therefore not inaccessible to ECA, but it is strongly conditioned by the IC family. Atlas rows should be read as claims about a stated rule/IC/scale protocol, not as intrinsic properties of the rule table alone.

9.5 PySR symbolic regression -- negative result

The decision-tree analyses (Section 4, temporal calibration) provide strong empirical signal but not closed-form symbolic expressions. PySR was planned as a follow-up to produce interpretable formulas for the calibrated thresholds and fragility spectra. PySR 1.5.10 was subsequently unblocked. A full regression over 15 atlas worlds (five features: `mean_laws`, `peak_diversity`, `noise_ratio`, `non_empty_ratio`, `f_core`; target `f_total`; 40 iterations) produced a best expression of complexity 9 with `MSE = 0.035`, above the threshold for a paper-worthy finding. The dominant predictor is `f_core`; the residual is driven by `rule_108` as a structural outlier (`f_gap = 0.945`) that is not predictable from aggregate features without a mechanism feature. The symbolic layer remains incomplete, and this negative result is consistent with the `f_core`/`f_gap` separation documented in Section 6.

10. Next Work

10.1 Symmetry-invariant observers

The two observer artifacts identified in Section 8 — `tipo_unico` mirror asymmetry and dedup translation non-equivariance — share a root cause: the heuristic observers do not encode the symmetries of the underlying CA. A natural next step is to build observers that canonicalize structure representations under spatial reflection and translation before counting. This would make `tipo_unico` a mirror-invariant physical property and dedup counts stable across IC positions, strengthening the evidential basis for both the atlas and the fragility measurements.

10.2 Extended local oscillator search

The `rule_108` uniqueness result holds under the current stationary protocol (zero background, exact period, IC words of length <= 12, span <= 32). Several controlled extensions have now been completed:

- **Moving oscillators**: completed. A companion sweep over all 128 quiescent rules found eight rules producing minimal period-2 speed-1 gliders (Section 7.5). No longer-period or slower-speed moving oscillators were found under this protocol. The follow-up length-9..12 sweep found no additional stationary or moving oscillator rules.
- **Longer IC words beyond 12**: extend the IC sweep from length 12 to length 16 or beyond to test whether substantially wider seed patterns produce oscillators in rules that failed the length-12 protocol.
- **Non-zero backgrounds**: completed. A sweep over 15 non-zero periodic backgrounds (template lengths 1, 2, 4) across all 256 rules and 502 IC words (1,927,680 runs) found that the periodic-background regime is substantially richer than the quiescent regime: 30 stationary rules, 36 moving rules, including period-4 oscillators and speed-0.5 gliders not present under zero background. `rule_108` persists under all-one background with the same motif. Full results in Section 7.6.
- **Period-8 backgrounds**: completed. A sweep over 30 primitive length-8 binary necklaces (3,855,360 runs) found 4 new stationary rules, 19 new moving rules, five new period classes (T=6, 8, 10, 12, 15), and a new glider speed (2/3 cell/step, T=3). All 10 sampled results were background-phase dependent after circular-geometry correction. Fase 25 confirms exact co-translation equivariance in 80/80 runs with circular shape canonicalization. Full results are in Section 7.7.
- **T=15 anatomy**: completed. All 221 detections belong to the reflection-symmetric, black/white-conjugate pair `rule_73/rule_109`. The 20 minimal rule/background representatives remain exact through step 900 and lock at five times the background temporal period (`15/3`). The basin is narrow: 23/160 background phases and 4/134 one-bit IC mutations retain `T=15`. Full results are in Section 7.8.
- **Five-state locking mechanism**: completed. The localized defect cycles through five distinct states under `F^3`. The ratio `T_local/T_bg=5` is the defect cycle length, confirmed in all 20 minimal representatives. Full results are in Section 7.9.
- **Induced defect algebra**: partial. The exact law `delta_f(b,d)=f(b XOR d) XOR f(b)` proves that the `rule_73` and `rule_109` defect orbits are identical under simultaneous black/white conjugation. Profiling all 100 `F^3` edges rejects a universal sparse-entry explanation and points instead to spatial phase or higher-order block states. Full results are in Section 7.10.
- **Block locality and shape families**: partial-positive. Fixed local defect blocks do not explain the T=15 cycle: the active defect already carries background-specific context at `W=0`, and no nontrivial shared block signature exists across all backgrounds. The variation is nevertheless finite: 20 representatives collapse into 13 phase-rotated shape families, with two families shared across the conjugate rules. Full results are in Section 7.11.
- **Compact T=15 state variable**: completed for the known representative set. No compact background-only descriptor predicts all 13 families globally, but the rule-conditioned descriptor `subpatterns_len4` separates all backgrounds per rule. Rotation generalization validates the compact state variable `(rule, subpatterns_len4, IC/background alignment)`: 140/140 co-translated rotations preserve both T=15 and the predicted family, while fixed-IC rotations collapse to 3/140 detections. Full results are in Section 7.12.

Each extension is a controlled experiment with the same measurement protocol; only the IC or background definition changes.

10.3 PySR symbolic regression

The decision-tree calibration for `frontera_temporal` and `temporal_scale_stability` (Section 4) provides thresholds but not formulas. PySR symbolic regression on the fragility spectrum (`f_total`, `f_core`, `f_gap` as functions of rule properties and IC metrics) is now technically available, but the first atlas-wide run did not produce a compact paper-worthy formula. The next useful symbolic-regression target would require mechanism features rather than aggregate atlas features alone.

10.4 Figures

The following six figures are planned for the preprint draft:

1. **World taxonomy table** — the full 20-world atlas with categories, law coverage symbols, and fragility columns, formatted as a paper-ready table.
2. **Law coverage matrix** — the `✓ / · / - / ?` matrix from `outputs/world_taxonomy/law_map.md`, rendered as a heatmap or binary grid.
3. **`f_total` / `f_core` spectrum** — a two-axis scatter or bar chart showing all measured worlds positioned by `f_total` and `f_core`, with the four fragility mechanisms labeled.
4. **`rule_108` oscillator motif** — a space-time diagram of the `#.# <-> ###` two-step cycle, showing several periods on a quiescent background.
5. **`rule_54` gate and observer non-equivariance** — a dual figure: the Fase 13 noise-gate crossing diagram (reference dedup vs perturbed dedup) alongside the Fase 19 per-position dedup variation (15..24 across k=0..63).
6. **Moving oscillator space-time diagram** -- dual panel showing `rule_20` (right-moving) and `rule_6` (left-moving) over 8 time steps, with active cells dark and quiescent background light. Illustrates the `[0] <-> [0,1]` glider cycle and the `+/-2` drift per period.

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